

1. Create a report that displays EMPNO, ENAME, DEPTNO of employees who work with SCOTT. (Nested Sub Query)

Query:

```
SELECT EMPNO, ENAME, DEPTNO
FROM EMP
WHERE DEPTNO=(SELECT DEPTNO FROM EMP WHERE ENAME='SCOTT')
```

Output:

```
SQL> SELECT EMPNO, ENAME, DEPTNO
2  FROM EMP
3  WHERE DEPTNO = (
4      SELECT DEPTNO
5      FROM EMP
6      WHERE ENAME = 'SCOTT'
7  );
```

EMPNO	ENAME	DEPTNO
7369	SMITH	20
7566	JONES	20
7788	SCOTT	20

2. Display the employees who earn more than the average salary of EMP table.

Query:

```
SELECT * FROM EMP WHERE SAL > (SELECT AVG(SAL) FROM EMP);
```

Output:

```
SQL> SELECT *
  2 FROM EMP
  3 WHERE SAL >
  4 (
  5     SELECT AVG(SAL)
  6     FROM EMP
  7 );
```

EMPNO	ENAME	JOB	MGR	HIREDATE
7566	JONES	MANAGER	7839	02-APR-81
2975		20		
7698	BLAKE	MANAGER	7839	01-MAY-81
2850		30		
7788	SCOTT	ANALYST	7566	19-APR-87
3000		20		

3. Display the ENAME, JOB who are Managers. (EXISTS)

Query:

```
SELECT E1.ENAME,E1.JOB FROM EMP E1
WHERE EXISTS (SELECT 1 FROM EMP E2 WHERE E2.MGR=E1.EMPNO);
```

Output:

```
SQL> SELECT E1.ENAME, E1.JOB
  2 FROM EMP E1
  3 WHERE EXISTS
  4 (
  5     SELECT 1
  6     FROM EMP E2
  7     WHERE E2.MGR = E1.EMPNO
  8 );
```

ENAME	JOB
BLAKE	MANAGER
JONES	MANAGER

4. Display the Employees who earn less than the least salary of DEPT 10. (ALL)

Query:

```
SELECT * FROM EMP
```

```
WHERE SAL < ALL (SELECT SAL FROM EMP WHERE DEPTNO=10);
```

Output;

```
8 );
```

EMPNO	ENAME	JOB	MGR	HIREDATE
7788	SCOTT	ANALYST	7566	19-APR-87
3000		20		
7566	JONES	MANAGER	7839	02-APR-81
2975		20		
7698	BLAKE	MANAGER	7839	01-MAY-81
2850		30		

EMPNO	ENAME	JOB	MGR	HIREDATE
7499	ALLEN	SALESMAN	7698	20-FEB-81
1600	300	30		
7521	WARD	SALESMAN	7698	22-FEB-81
1250	500	30		
7654	MARTIN	SALESMAN	7698	28-SEP-81
1250	1400	30		

EMPNO	ENAME	JOB	MGR	HIREDATE
7369	SMITH	CLERK	7902	17-DEC-80
800		20		

5. Display the employees who have the same DEPTNO and MGR as SCOTT excluding SCOTT.

Query:

```
SELECT * FROM EMP
WHERE (DEPTNO,MGR)=(SELECT DEPTNO,MGR FROM EMP WHERE
ENAME='SCOTT')
AND ENAME<>'SCOTT';
```

Output:

```
SQL> SELECT *
  2  FROM EMP
  3  WHERE (DEPTNO, MGR) =
  4  (
  5      SELECT DEPTNO, MGR
  6      FROM EMP
  7      WHERE ENAME='SCOTT'
  8  )
  9  AND ENAME <> 'SCOTT';

no rows selected
```

6. Write a query that displays the employee number and name of all employees who work in a department with any employee whose name contains R.

Query:

```
SELECT EMPNO,ENAME FROM EMP
WHERE DEPTNO IN (SELECT DEPTNO FROM EMP WHERE ENAME LIKE '%R%');
```

Output:

```
SQL> SELECT EMPNO, ENAME
  2  FROM EMP
  3  WHERE DEPTNO IN
  4  (
  5      SELECT DEPTNO
  6      FROM EMP
  7      WHERE ENAME LIKE '%R%'
  8  );
```

EMPNO	ENAME
7499	ALLEN
7521	WARD
7654	MARTIN
7698	BLAKE

7. The HR department needs a report that displays the employee ENAME, DEPTNO, JOB of all employees whose work in NEW YORK.

Query:

```
SELECT ENAME,DEPTNO,JOB FROM EMP
WHERE DEPTNO=(SELECT DEPTNO FROM DEPT WHERE LOC='NEW YORK');
```

Output;

```
SQL> SELECT ENAME, DEPTNO, JOB
  2  FROM EMP
  3  WHERE DEPTNO =
  4  (
  5      SELECT DEPTNO
  6      FROM DEPT
  7      WHERE LOC='NEW YORK'
  8  );
```

no rows selected

8. Modify the query so that the user is prompted for LOC.

Query:

```
ACCEPT LOCNAME CHAR PROMPT 'Enter Location:'  
SELECT ENAME,DEPTNO,JOB FROM EMP  
WHERE DEPTNO=(SELECT DEPTNO FROM DEPT WHERE LOC='&LOCNAME');
```

Output:

```
SQL> ACCEPT LOCNAME CHAR PROMPT 'Enter Location: '  
Enter Location: DALLAS  
SQL> SELECT ENAME, DEPTNO, JOB  
2 FROM EMP  
3 WHERE DEPTNO =  
4 (  
5     SELECT DEPTNO  
6     FROM DEPT  
7     WHERE LOC='&LOCNAME '  
8 );  
old 7: WHERE LOC='&LOCNAME '  
new 7: WHERE LOC='DALLAS '  
  
ENAME                DEPTNO JOB  
-----  
SMITH                20 CLERK  
JONES                20 MANAGER  
SCOTT                20 ANALYST
```

9. create a report for HR that displays the Name and salary of every employee reports to KING.

Query:

```
SELECT ENAME,SAL FROM EMP
WHERE MGR=(SELECT EMPNO FROM EMP WHERE ENAME='KING');
```

Output:

```
SQL> SELECT ENAME, SAL
  2  FROM EMP
  3  WHERE MGR =
  4  (
  5      SELECT EMPNO
  6      FROM EMP
  7      WHERE ENAME='KING'
  8  );

no rows selected
```

10. Write a query to display all employees working with JAMES.

Query:

```
SELECT * FROM EMP
WHERE DEPTNO=(SELECT DEPTNO FROM EMP WHERE ENAME='JAMES')
AND ENAME<>'JAMES';
```

Output:

```
SQL> SELECT *
  2  FROM EMP
  3  WHERE DEPTNO =
  4  (
  5      SELECT DEPTNO
  6      FROM EMP
  7      WHERE ENAME='JAMES'
  8  )
  9  AND ENAME <> 'JAMES';

no rows selected
```

11. Displays all the employees who earn less than average salaries of their respective department.

Query:

```
SELECT * FROM EMP E
WHERE SAL < (SELECT AVG(SAL) FROM EMP WHERE DEPTNO = E.DEPTNO);
```

Output:

```
SQL> SELECT *
2  FROM EMP E
3  WHERE SAL <
4  (
5      SELECT AVG(SAL)
6      FROM EMP
7      WHERE DEPTNO = E.DEPTNO
8  );
```

EMPNO	ENAME	JOB	MGR	HIREDATE
7369	SMITH	CLERK	7902	17-DEC-80
800		20		
7499	ALLEN	SALESMAN	7698	20-FEB-81
1600	300	30		
7521	WARD	SALESMAN	7698	22-FEB-81
1250	500	30		

EMPNO	ENAME	JOB	MGR	HIREDATE
7654	MARTIN	SALESMAN	7698	28-SEP-81
1250	1400	30		

12. Write a query to display LOC and average salary of each location.

Query:

```
SELECT D.LOC,(SELECT AVG(SAL) FROM EMP E WHERE E.DEPTNO=D.DEPTNO)
AVG_SAL
FROM DEPT D;
```

Output:

```
SQL> SELECT D.LOC,
2  (
3      SELECT AVG(SAL)
4      FROM EMP E
5      WHERE E.DEPTNO = D.DEPTNO
6  ) AS AVG_SAL
7  FROM DEPT D;
```

LOC	AVG_SAL
-----	-----
DALLAS	2258.33333
CHICAGO	1737.5
BOSTON	
NEW YORK	

13. Write a query to display least N salaries. Use In-Line views .

Query:

```
SELECT * FROM (SELECT * FROM EMP ORDER BY SAL)
WHERE ROWNUM<=3;
```

Output:

```
SQL> SELECT *
  2  FROM
  3  (
  4      SELECT *
  5      FROM EMP
  6      ORDER BY SAL
  7  )
  8  WHERE ROWNUM <= 3;
```

EMPNO	ENAME	JOB	MGR	HIREDATE
7369	SMITH	CLERK	7902	17-DEC-80
7521	WARD	SALESMAN	7698	22-FEB-81
7654	MARTIN	SALESMAN	7698	28-SEP-81

14. Display last N rows from the employees table(Use Correlated Sub Queries).

Query:

```
SELECT * FROM EMP E1
WHERE 3>(SELECT COUNT(*) FROM EMP E2 WHERE E2.EMPNO>E1.EMPNO);
```

Output:

```
SQL> SELECT *
2  FROM EMP E1
3  WHERE 3 >
4  (
5      SELECT COUNT(*)
6      FROM EMP E2
7      WHERE E2.EMPNO > E1.EMPNO
8  );
```

EMPNO	ENAME	JOB	MGR	HIREDATE
SAL	COMM	DEPTNO		
7654	MARTIN	SALESMAN	7698	28-SEP-81
1250	1400	30		
7698	BLAKE	MANAGER	7839	01-MAY-81
2850		30		
7788	SCOTT	ANALYST	7566	19-APR-87
3000		20		

15. Display employees from employees table and sort only employees working in DALLAS.(Scalar Sub Query)

Query:

```
SELECT * FROM EMP
WHERE DEPTNO=(SELECT DEPTNO FROM DEPT WHERE LOC='DALLAS')
ORDER BY ENAME;
```

Output:

```
SQL> SELECT *
2  FROM EMP
3  WHERE DEPTNO =
4  (
5      SELECT DEPTNO
6      FROM DEPT
7      WHERE LOC='DALLAS'
8  );
```

EMPNO	ENAME	JOB	MGR	HIREDATE
7369	SMITH	CLERK	7902	17-DEC-80
800		20		
7566	JONES	MANAGER	7839	02-APR-81
2975		20		
7788	SCOTT	ANALYST	7566	19-APR-87
3000		20		

16. Display the employees who earn a salary less than avgsal of their respective department. Also display the avgsal.(Use Inline Views)

Query:

```
SELECT E.EMPNO,E.ENAME,E.SAL,A.AVGSAL
FROM EMP E,
(SELECT DEPTNO,AVG(SAL) AVGSAL FROM EMP GROUP BY DEPTNO) A
WHERE E.DEPTNO=A.DEPTNO AND E.SAL<A.AVGSAL;
```

Output:

```
SQL> SELECT E.EMPNO,
2      E.ENAME,
3      E.SAL,
4      A.AVGSAL
5 FROM EMP E,
6 (
7     SELECT DEPTNO,
8            AVG(SAL) AVGSAL
9     FROM EMP
10    GROUP BY DEPTNO
11 ) A
12 WHERE E.DEPTNO = A.DEPTNO
13 AND E.SAL < A.AVGSAL;
```

EMPNO	ENAME	SAL	AVGSAL
7369	SMITH	800	2258.33333
7499	ALLEN	1600	1737.5
7521	WARD	1250	1737.5
7654	MARTIN	1250	1737.5

17. Write a query that display LOC of those DEPTS that have a sum of sal less than the avg sal of all the employees in DEPT table.(Use WITH Clause)

Query:

```
WITH DEPTSAL AS (  
SELECT DEPTNO,SUM(SAL) TOTALSAL FROM EMP GROUP BY DEPTNO),  
AVGEMP AS (  
SELECT AVG(SAL) AVGSAL FROM EMP)  
SELECT D.LOC  
FROM DEPT D JOIN DEPTSAL DS ON D.DEPTNO=DS.DEPTNO  
CROSS JOIN AVGEMP A  
WHERE DS.TOTALSAL<A.AVGSAL;
```

Output:

```
SQL> WITH DEPTSAL AS  
2  (  
3      SELECT DEPTNO,  
4          SUM(SAL) TOTALSAL  
5      FROM EMP  
6      GROUP BY DEPTNO  
7  ),  
8  AVGEMP AS  
9  (  
10     SELECT AVG(SAL) AVGSAL  
11     FROM EMP  
12  )  
13  SELECT D.LOC  
14  FROM DEPT D  
15  JOIN DEPTSAL DS  
16  ON D.DEPTNO = DS.DEPTNO  
17  CROSS JOIN AVGEMP A  
18  WHERE DS.TOTALSAL < A.AVGSAL;  
  
no rows selected
```